

TITLE : LASER SECURITY SYSTEM

ABSTRACT

The Laser Security System is a modern, advanced solution designed to provide precise and reliable security by creating an invisible perimeter around a designated area using laser beams. The system utilizes infrared or visible lasers to continuously monitor the protected space, detecting any intrusion when a laser beam is interrupted. This technology offers a high level of accuracy, minimizing false alarms that are often common with traditional motion detectors.

The system can be customized to suit various environments, with adjustable beam ranges and sensitivity settings. It is ideal for residential, commercial, and high-security applications, offering an efficient and non-intrusive way to secure properties. Additionally, the Laser Security System is designed for seamless integration with existing security infrastructures, such as surveillance cameras or alarms, enhancing overall security monitoring.

Built with weather-resistant materials, the system is durable and reliable in diverse environmental conditions, ensuring year-round protection. With low power consumption and flexible installation options, including both wired and battery-powered configurations, the Laser Security System provides an energy-efficient, cost-effective solution for modern perimeter security needs.

FIELD OF INVENTION

The invention pertains to advancements in the field of security technology, specifically focusing on protective systems for buildings, properties, and other sensitive areas. It integrates laser-based sensors and detection systems to create a highly effective, non-intrusive method of safeguarding environments against unauthorized entry.

BACKGROUND OF THE INVENTION

As security concerns grow, there is a need for more efficient and reliable systems. Traditional alarm systems often suffer from false triggers due to non-threatening movements, while physical barriers like fences can be costly and require maintenance. The Laser Security System overcomes these challenges by using infrared or visible laser beams to create an invisible perimeter. When disrupted by an intruder, the system accurately detects the breach, providing a precise and nearly impenetrable security solution suitable for both residential and commercial applications.

SUMMARY OF THE INVENTION

The Laser Security System offers a novel approach to property protection by using infrared or visible laser beams to create an invisible perimeter. When disrupted by an intruder, the system immediately detects the breach and triggers a response. It integrates seamlessly with existing security systems, is highly customizable for different environments, and features a weather-resistant design for reliable year-round use. Additionally, the system is energy-efficient, with both wired and wireless installation options, providing a flexible and reliable security solution for various applications.

SPECIFICATIONS

Material:

- **Laser Sensors:** High-quality laser diodes and photodetectors for accurate beam transmission and reception.
- **Casing:** Weather-resistant plastic or metal casing, designed to protect the sensors from external elements.
- **Alarm System:** Wired or wireless alarm system with a high-decibel siren and a built-in notification feature to alert connected devices.

Dimensions:

- **Sensor Range:** Up to 30 meters for outdoor applications, adjustable for smaller indoor environments.
- **Laser Beam Width:** Adjustable from 1 cm to 5 cm for various levels of sensitivity.

Power Supply:

- **Battery Operated:** Rechargeable battery or wired for continuous power supply.
- **Power Consumption:** Low energy consumption, designed for 24/7 operation with minimal energy usage.

Detection Mechanism:

- **Beam Type:** Infrared laser beams, invisible to the human eye.
- **Response Time:** 0.5 seconds from beam interruption to alarm activation.
- **Alarm Output:** 110 dB siren.

Weight:

- **Sensor Unit:** 200 grams each (lightweight for easy installation).
- **Control Unit:** 1.5 kg (if separate from the sensors).

DESCRIPTION

The Laser Security System operates by emitting a series of invisible infrared laser beams that create a secure perimeter around the designated area. These beams are arranged in a specific pattern depending on the security needs, whether it's in a straight line or a grid layout. When an object, such as a person or animal, interrupts the laser path, the system detects the breach in real-time.

Upon detecting an intrusion, the system can activate a local alert such as an indicator light or a mechanical signal to notify the user or security personnel on-site. The laser beams are safe to use and reliable in any lighting conditions, ensuring optimal security both during the day and at night.

The Laser Security System is easy to install and can be tailored to suit a variety of environments, from residential properties to high-security commercial sites. The system's flexibility and robustness make it an excellent choice for enhancing security in any location.



Figure 1



Figure 2

WE CLAIM

1. **Claim:** The Laser Security System provides an invisible, highly precise security perimeter using infrared or visible laser beams to detect unauthorized entry.
2. **Claim:** The system includes an automatic alarm that is triggered when the laser beams are disrupted by an intruder.
3. **Claim:** The Laser Security System is customizable in terms of beam range and sensitivity to accommodate a wide variety of security needs.
4. **Claim:** The system is designed with durable, weather-resistant components, ensuring reliable performance in diverse environmental conditions.
5. **Claim:** The laser sensors feature low power consumption, long battery life, and can be connected to external devices for monitoring and notification.

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